NOCTUO

PRODUCT CARD

HIPS V0

flame retardant properties

high stifness

good heat resistance

high impact resistance

1. GENERAL INFORMATION ABOUT THE PRODUCT

A specialist variety of high-impact polystyrene, characterized by a high degree of flame retardancy, allowing it to meet the requirements for the V0 class according to UL94 standards, while maintaining good printability and good mechanical parameters.

Main HIPS V0 features:

- complies with the V0 flammability criteria according to the UL-94 standard,
- good mechanical strength,
- high impact and cracks resistance,
- good thermal stability.

2. TECHNICAL PARAMETERS

CHARACTERISTICS	TEST METHOD	TEST CONDITIONS	IU	VALUE
	ISO			
PHYSICAL				
Density	ISO 1183	-	g/cm^3	1.14
Water absorption	ISO 60	23°C/24h	%	<0.1
MECHANICAL				
Yield strength	ISO 527-1,-2	50mm/min	MPa	23
Elongation at break	ISO 527-1,-2	50mm/min	%	50
Elastic modulus at tension	ISO 527-1,-2	1mm/min	MPa	1600
Bending strenght	ISO 178	2mm/min	MPa	32
Notched impact strength, IZOD	ISO 180/4A	23°C-3,2mm	J/m	81

Notched impact strength, IZOD	ISO 180/1A	23°C - 4mm	kJ/m^2	7
THERMAL				
Vicat softening point	ISO 306	50N	°C	90
Deflection temperature under load	ASTM D 648	1,8 MPa	°C	85
Coefficient of linear thermal expansion II/⊥	ASTM D 696	23°C - 85°C	E-6/°C	90
COMBUSTIBILITY				
Flammability level test	UL94	1,5 mm	Class	V0

Tests have been done in 23°C, if it's not marked differently.

3. RECOMMENDATION OF PRINTING

There might be some problems to achieve proper adhesion on glass bed.
Using perforated bed, PVA glue, specialized product (adapted to work with ABS and derivatives), kapton coating or other different way to obtain adhesion (for example 'ABS juice' or hair spray) is recommended.
Similar to other styrene filaments, it is good practice to be slow receiving temperature from freshly applied layers – closed chamber and minimal or turned off the cooling are recommended.
It is necessary to use only with fume extraction or in very well ventilated conditions. In case of prolonged exposure of the filament to high moisture, drying may be necessary (recommended drying temperature 70 °C).

Recommended parameters of printing:

Hotend temperature	230 - 260 °C
Bed temperature	> 80 °C
Print speed	< 100 mm/s

Examples of problems and their solutions:

Problem	Possible cause	Proposed solution
Weak layers adhesion	1) Extruder temperature is too low 2) Too high speed of printing 3) Too efficient cooling	1) Raising the extruder temperature 2) Raising the extruder temperature / reducing the print speed 3) Reducing or turning off the fan responsible for cooling the print
Uneven feeding – losing steps of the feeder / filament sliding on the drive gear	1) Extruder temperature is too low 2) Weak feader pressure	1) Raising the extruder temperature 2) Raising the pressure
Print is peeling of the bed	1) Too low temperature of bed 2) Improperly prepared surface 3) Cooling	1) Raising the bed temperature 2) Degreasing the bed / using another source of adhesion / using perforated bed 3) Giving up the cooling on first stage is recommended
Edge curling	1) Processing shrinkage 2) Too high acceleration	1) Compensation with amount of served filament / selection cooling parameters / changing the chamber temperature 2) Due to high stickiness, high acceleration with sharp edges may cause pulling the edges up

4. SAFETY NOTES

Exhaust fan is recommended.
Air filters in printer is recommended.
HIPS V0 needs to be used only in well ventilated conditions.
Inhaling fumes generated during the printing must be avoided.

Generating fumes during the printing depends mainly on printing temperature. In case of visibly raising emission level, the printing needs to end. Check the hotend temperature and efficiency of the control system before using it next time.

It's forbidden to set fire or exceed recommended printing temperature!

Decomposition of HIPS V0 is typically over 300 °C.
Main ingredient of decomposition is styrene. Printing at too high temperature can cause excessive release of flame retardant fumes. Although the product does not contain PBBE flame retardants, use due care and personal protective equipment when in use.

Detailed safety information available in SDS.